

The Internet of Humanoids: A Survey of Technologies, Applications, and Challenges

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Abstract—Humanoid robots are increasingly capable of operating autonomously in human-centric environments, motivating the emergence of the Internet of Humanoids (IoH)—a paradigm that interconnects humanoid robots into collaborative, intelligent networks. Unlike traditional Internet of Things (IoT) systems that link passive or task-specific devices, IoH treats humanoids as embodied, autonomous, and safety-critical agents with tightly coupled perception, decision-making, communication, and actuation. Leveraging advances in large multimodal models (LMMs), cloud-edge computing, and next-generation wireless connectivity, IoH enables real-time coordination, collective learning, and scalable fleet deployment across domains such as healthcare, manufacturing, logistics, and disaster response. This article synthesizes the IoH landscape, proposes a layered architectural reference model to unify heterogeneous subsystems, and critically analyzes challenges in security, privacy, scalability, interoperability, and ethics. We further outline open research directions toward secure, trustworthy, and interoperable humanoid networks at scale.

Index Terms—Agents, artificial intelligence (AI), humanoids, Internet of Humanoids (IoH).

I. INTRODUCTION

HUMANOID robots have rapidly emerged as a major research and development frontier due to their ability to operate autonomously in human-centric environments such as homes, hospitals, factories, and public spaces [1]. Recent advances in artificial intelligence (AI), wireless networking, and distributed systems are further transforming humanoids from standalone machines into networked, collaborative agents.

This convergence motivates the Internet of Humanoids (IoH), a new paradigm in which autonomous humanoid robots function as interconnected nodes within a shared computational and communication ecosystem [2], [3], [4]. In IoH (see Fig. 1), humanoids collaborate, share knowledge, and coordinate tasks across domains including healthcare, domestic service, smart manufacturing, logistics, and disaster response. While IoH builds upon foundational concepts from the Internet of Things (IoT), it differs fundamentally

in scope; IoH centers on embodied, autonomous, and safety-critical agents that tightly couple perception, decision-making, communication, and physical actuation [5], [6]. By integrating embodied autonomy with advanced AI and next-generation communication technologies such as 6G, IoH represents a shift from networked devices to networked embodied intelligence.

The growing momentum behind IoH is driven by several converging factors. Advances in humanoid hardware and control—exemplified by platforms such as Digit, Figure 02, and Optimus—have enabled robust locomotion and manipulation beyond laboratory settings. In parallel, mature cloud and IoT infrastructures provide scalable backbones for distributed robotic systems [7], while emerging 6G technologies promise ultralow latency and high reliability for time-critical humanoid applications [8], [9], [10]. Societal pressures, including aging populations, labor shortages, and increasing demand for care and service automation, further motivate the deployment of IoH at scale [11].

Despite this progress, existing surveys typically examine humanoid robotics, cloud robotics, or IoT systems in isolation. They do not adequately address the cross-layer challenges that arise when humanoid robots operate as large-scale, autonomous, and collaborative networks. In particular, IoH introduces significant challenges related to security and privacy, scalability and interoperability across heterogeneous platforms, computational and energy constraints, and ethical concerns surrounding trust, accountability, and human-robot coexistence [12].

Motivated by these observations, this article provides a focused and comprehensive survey of IoH, emphasizing its architectural foundations, enabling technologies, application domains, and open research challenges. The main contributions of this article are summarized as follows.

- 1) We present the conceptual background and system context necessary for understanding the IoH paradigm.
- 2) We review key enabling technologies supporting large-scale humanoid connectivity and intelligence.
- 3) We survey representative application domains where interconnected humanoids deliver societal value.
- 4) We analyze major technical, ethical, and system-level challenges facing IoH deployments.
- 5) We identify forward-looking research directions toward secure, scalable, and interoperable IoH ecosystems.

The remainder of this article is organized as follows. Section II introduces background concepts. Section III reviews enabling technologies. Section IV surveys applications. Sec-

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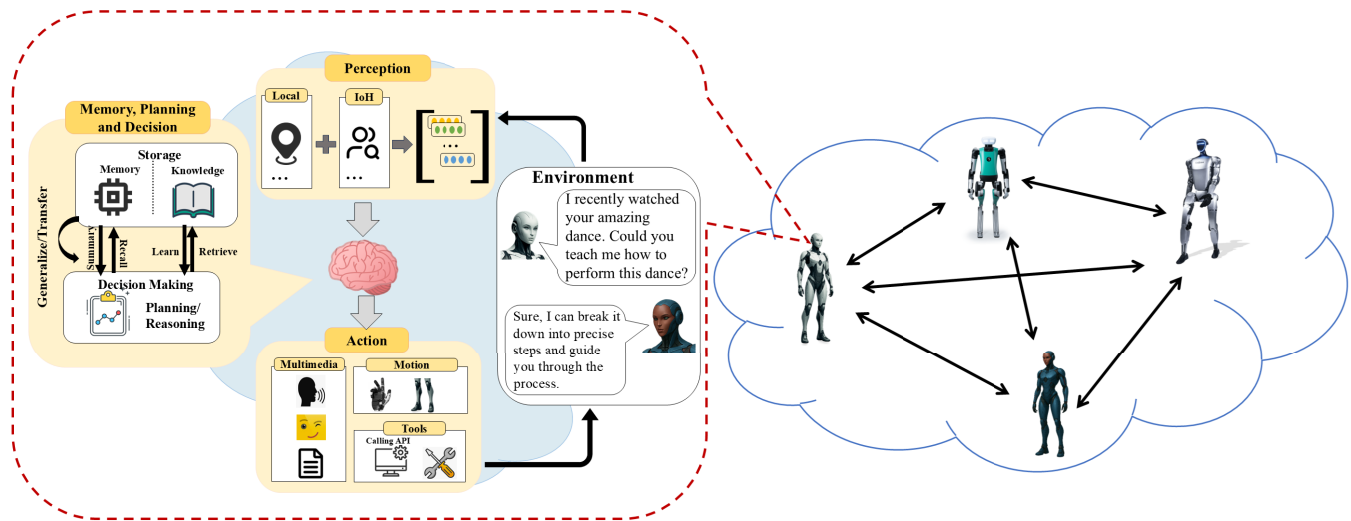


Fig. 1. IoH.

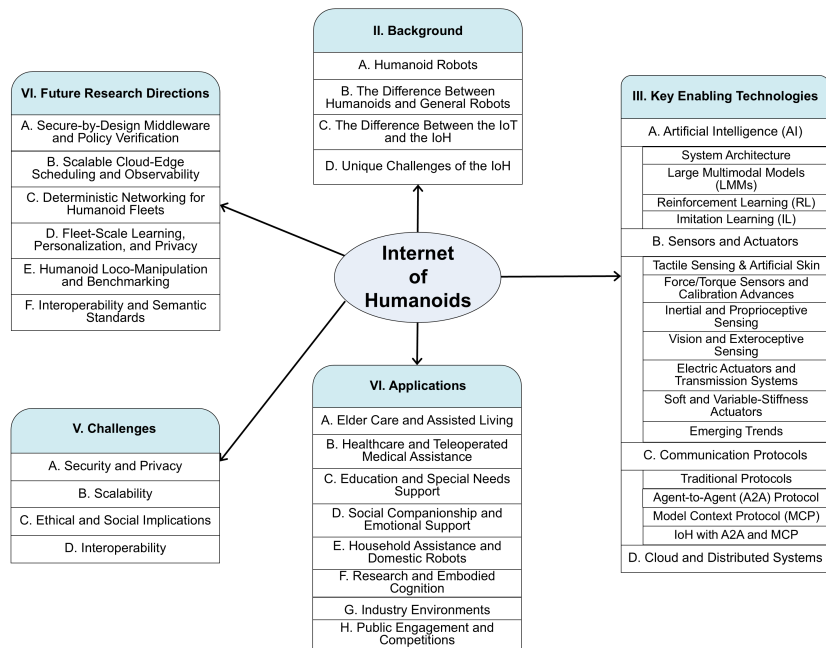


Fig. 2. Road map of this article.

tion V discusses challenges, and Section VI outlines future research directions before concluding in Section VII. The overall structure is illustrated in Fig. 2.

II. BACKGROUND

This section begins by introducing humanoid robots and contrasting them with general-purpose robotic systems. It then discusses the differences between IoT and IoH and outlines the unique challenges associated with IoH systems.

A. Humanoid Robots

Humanoid robots are sophisticated robotic systems engineered to closely emulate human appearance, movement, and behavior. They integrate advanced capabilities such as bipedal

locomotion, dexterous manipulation, and multimodal sensory systems that approximate human-like perception [13]. Early milestones in humanoid robotics include WABOT-1 (1973) and Honda’s ASIMO (2000), which demonstrated foundational abilities in locomotion, balance, and basic human–robot interaction [5].

These robots are designed with anthropomorphic mechanical structures that mirror the human body, including articulated limbs, torso configurations, and joint layouts. This human-inspired morphology enables humanoids to operate effectively in environments designed for people, supporting tasks that require whole-body coordination, reachability, and dexterity. Their sensory systems—typically comprising vision (e.g., stereo or RGB-D cameras), auditory processing (e.g., microphone arrays), and tactile feedback (e.g., pressure-

or force-sensitive surfaces)—allow humanoids to perceive, interpret, and respond to complex environmental stimuli in a manner analogous to human sensory processing. As a result, humanoids can interact more intuitively with both objects and people, adapting to dynamic and unstructured settings.

Humanoid robotics has advanced rapidly in recent years. The global market is projected to grow from U.S. 2.03 billion in 2024 to U.S. 13.25 billion by 2029, driven by progress in actuation, sensing, AI, and large-scale learning [14]. Leading companies such as Boston Dynamics, Tesla, Figure AI, and Unitree have introduced state-of-the-art platforms, including Atlas, Optimus, Figure 02, and G1. Boston Dynamics' Atlas is widely recognized for its highly dynamic locomotion capabilities, including parkour and backflips [15]. Tesla's Optimus emphasizes general-purpose manipulation supported by large-scale AI, Figure 02 focuses on safe and efficient human–robot collaboration, and Unitree's G1 targets affordability and versatility, lowering barriers to deployment [16]. Together, these platforms illustrate a shift from laboratory prototypes toward practical, real-world humanoid deployment.

Additional humanoid platforms further highlight the diversity of applications. Ameca, developed by Engineered Arts, excels in expressive social interaction through highly articulated facial expressions, indicating strong potential for customer service and entertainment. Digit, developed by Agility Robotics, targets logistics and warehouse automation, navigating complex environments while carrying payloads of up to 15 kg. Apollo by Apptронik supports industrial automation and elder care, handling payloads up to 25 kg with advanced safety features [17]. ALTER 3, a collaboration between Osaka University and Mixi Corporation, integrates humanoid robotics with artistic performance, including orchestral conducting and theatrical expression [18].

These developments collectively demonstrate the transformative potential of humanoid robotics across logistics, healthcare, industry, and creative domains. At the same time, significant challenges remain, including high system costs, limited energy efficiency, and the difficulty of sustaining long-duration autonomous operation in real-world environments [14].

B. Difference Between Humanoids and General Robots

Unlike general-purpose robots—such as industrial robotic arms commonly used in manufacturing or wheeled platforms deployed in logistics—humanoid robots are purposefully designed to operate within human-centric environments, including homes, hospitals, offices, and public spaces [19]. These environments impose fundamentally different requirements on robotic systems. Humanoids must maintain dynamic balance while navigating uneven floors, stairs, and cluttered hallways, and must adapt their behavior to continuously changing human activity and social context. In contrast, most general robots operate in structured or semi-structured settings optimized for automation and repeatability.

Beyond locomotion and manipulation, humanoid robots are often equipped with explicit social interaction capabilities. These capabilities enable humanoids to interpret human

TABLE I
COMPARISON OF HUMANOIDS AND GENERAL ROBOTS

Aspect	Humanoids	General Robots
Appearance	Human-like morphology (bipedal locomotion, head, torso, arms, legs)	Task-specific designs (e.g., robotic arms, wheeled platforms, drones)
Primary Purpose	Human–robot interaction, assistance, and operation in human-centric environments	Task automation and efficiency in structured or controlled settings
System Complexity	High, due to whole-body coordination, balance control, and social interaction requirements	Lower, focused on functional simplicity and task optimization
Application Domains	Social assistance, caregiving, service, research (e.g., Atlas)	Industrial automation, logistics, cleaning, exploration
AI Emphasis	Cognitive reasoning, social intelligence, perception–action coupling	Control, precision, optimization, and task execution
Cost	Generally higher due to complex mechanics and sensing	Varies, often lower for specialized platforms

intent, recognize facial expressions and gestures, and respond appropriately to verbal and nonverbal cues. These functions are largely unnecessary for task-specific robots. Such social awareness is essential for safe and effective deployment in shared human spaces.

For example, in a hospital setting, a humanoid robot may support healthcare workers by delivering medical supplies, monitoring patients, or providing companionship to individuals in long-term care. These tasks require not only physical dexterity and mobility, but also contextual awareness and empathetic interaction. The anthropomorphic form factor of humanoid robots facilitates intuitive human–robot interaction and allows them to seamlessly integrate into infrastructures originally designed for people, thereby reducing the need for costly environmental modifications.

Table I provides a comparative overview of humanoid robots and general-purpose robots. Key distinctions are highlighted in terms of morphology, functional objectives, system complexity, application domains, and AI emphasis. This comparison underscores why humanoids represent a distinct class of robotic systems whose networking, coordination, and safety requirements motivate IoH paradigm discussed later in this article.

C. Difference Between the IoT and the IoH

IoT refers to a networked ecosystem that interconnects physical devices, such as sensors, appliances, and smart-home systems, to enable seamless data exchange and automation [7]. These devices, ranging from temperature sensors to smart thermostats, communicate through standardized protocols like Message Queuing Telemetry Transport (MQTT) or constrained application protocol (CoAP). Real-time data collection, monitoring, and remote control are facilitated. IoT systems are designed to enhance efficiency in applications such as smart homes, industrial automation, and environmental monitoring. They primarily focus on passive data aggregation and predefined automation tasks. For instance, an IoT-enabled

TABLE II
COMPARISON OF THE IoT, CLOUD ROBOTICS, NETWORKED ROBOTICS, AND IOH

Aspect	IoT	Cloud Robotics	Networked Robotics	Internet of Humanoids (IoH)
Primary entities	Sensors, actuators, smart devices	Robots with cloud offloading	Multiple robots coordinating tasks	Humanoid robots with human-like bodies and cognition
Embodiment	Minimal or none	Often wheeled or fixed-base robots	Heterogeneous robots (often non-humanoid)	High-DoF humanoid embodiment with bipedal locomotion and whole-body manipulation
Perception–Action Coupling	Weak, event-driven	Moderate, partially cloud-mediated	Moderate, task-level coordination	Tightly coupled real-time perception–decision–action loops
Latency Tolerance	High (ms–seconds acceptable)	Moderate (cloud delays tolerated)	Moderate (task-level delays tolerated)	Ultra-low latency required for balance, contact, and safety
Safety Criticality	Low	Medium	Medium	High: failures can cause physical harm to humans
Cognition and Semantics	Rule-based	Centralized planning and learning	Shared task-level representations	Distributed cognition and semantic grounding
Human Interaction	Indirect	Limited	Structured environments	Continuous physical and social interaction
Autonomy Model	Centralized or scripted	Cloud-assisted autonomy	Coordinated multi-robot autonomy	Hybrid local autonomy with network-level coordination
Failure Consequences	Reduced efficiency	Task interruption	Coordination inefficiency	Falls, collisions, unsafe force, loss of trust
Key Architectural Requirement	Connectivity and scalability	Compute offloading	Task synchronization	Embodied real-time coordination, semantic consistency, safety guarantees

thermostat can collect temperature data and adjust heating based on predefined rules, requiring minimal human intervention.

From an architectural perspective, IoT nodes are typically characterized by limited computation, static or quasi-static deployment, weak coupling between sensing and actuation, and tolerance to latency or packet loss. Control decisions are often centralized or rule-based, and failures usually degrade system performance rather than causing immediate physical hazards.

IoH extends beyond the IoT paradigm by incorporating humanoid robots as autonomous, intelligent, and mobile nodes within a networked ecosystem, as illustrated in Fig. 1. Unlike traditional IoT devices, humanoids possess high-dimensional articulated bodies, nonholonomic locomotion constraints, continuous force interaction with the environment, and tight real-time coupling between perception, cognition, and actuation. This enables them to perform complex physical tasks in human-centric environments, such as caregiving, collaborative manufacturing, and public service [1], [2].

Critically, IoH cannot be viewed as a simple extension of cloud robotics or networked robotics. While cloud robotics focuses on offloading computation (e.g., mapping, planning, and learning) and networked robotics emphasizes coordination among multiple robots, IoH explicitly targets *humanoid embodiment*—where bipedal locomotion, whole-body balance, dexterous manipulation, and social interaction impose strict real-time and safety constraints that are absent in wheeled robots or industrial manipulators. For example, a humanoid’s balance controller may require submillisecond feedback loops and consistent global state estimates; delays or inconsistencies that are acceptable in cloud robotics can lead to falls or unsafe contact in humanoid systems.

To clearly position IoH with respect to adjacent paradigms, Table II provides a concise architectural comparison between the IoT, cloud robotics, networked robotics, and the IoH.

As a result, the IoH introduces fundamentally different architectural requirements. While IoT systems prioritize data-driven automation and throughput, the IoH emphasizes *embodied real-time coordination*, semantic understanding, and safety-aware decision-making. Humanoids must continuously interpret multimodal sensory input, reason about physical and social context, and adapt actions under uncertainty—all while coordinating with other humanoids and humans. This necessitates ultralow-latency communication, local autonomy with network-aware fallback, and semantic-level coordination mechanisms that go beyond existing IoT or cloud-robotics frameworks [1].

For instance, in a hospital setting, an IoH-enabled humanoid must not only coordinate task allocation (e.g., supply delivery) with peers but also maintain balance on uneven floors, yield to human motion, and adapt force during physical interaction—capabilities that require tight integration of networking, control, and cognition rather than simple message passing.

D. Unique Challenges of the IoH

IoH inherits scalable networking paradigms from IoT, such as cloud computing and edge processing, but introduces a set of qualitatively different and nontrivial challenges that stem from humanoids being embodied, cognitive, and socially situated agents, rather than abstract computational nodes. Unlike traditional IoT nodes or even generic robots, humanoid robots tightly couple perception, high-level reasoning, and physical action through high-dimensional kinematic chains and whole-body dynamics. As a result, failures in communication, synchronization, or decision-making can immediately

manifest as unsafe physical behavior, making IoH systems fundamentally *cyber-physical-cognitive* in nature.

A first distinctive challenge lies in embodied real-time coordination under humanoid kinematic and dynamic constraints. Real-time synchronization among humanoids is required not only to avoid logical conflicts (e.g., redundant task execution) but also to ensure stability and safety during whole-body locomotion and manipulation. Recent work on humanoid loco-manipulation shows that even millisecond-scale communication delays or state estimation inconsistencies can destabilize gaits, misalign contacts, or cause unsafe force application during collaborative tasks [20], [21]. Such tight coupling between networking, control, and embodiment is largely absent in IoT systems and significantly more demanding than in wheeled or fixed-base robots.

A second nonobvious challenge is semantic and cognitive interoperability at scale. In IoH, humanoids must coordinate not only low-level sensor data but also high-level abstractions such as task goals, affordances, human intent, and social norms. Vision-language-action (VLA) and multimodal foundation models have shown that different humanoids—trained on heterogeneous data and architectures—may form divergent internal representations of identical instructions or environments [22]. Without shared semantic grounding, such discrepancies can lead to inconsistent or unsafe collective behavior, revealing a gap that existing IoT or networked-robotics interoperability solutions do not address.

A third challenge arises from distributed autonomy, responsibility, and explainability. In IoH, humanoids must decide when to act independently and when to rely on network consensus while operating under safety, ethical, and legal constraints. Assigning responsibility and providing explanations for failures or near-misses becomes particularly complex when decisions emerge from interactions between local policies, shared models, and networked coordination. These issues are especially critical in healthcare, elder care, and public environments, where accountability and transparency are essential.

These challenges are further amplified in human-robot collaborative environments. A humanoid assisting a human must continuously infer human motion and intent while coordinating with peer humanoids over the network. This requires fault-tolerant, ultralow-latency communication combined with robust local fallback behaviors, ensuring safety even under partial network degradation or disconnection.

In addition, ethical and social challenges in IoH are qualitatively distinct from those in IoT. Humanoids collect rich multimodal data—including video, audio, posture, and behavioral cues—within intimate human spaces, increasing risks related to privacy leakage, emotional dependency, long-term trust calibration, and informed consent. Addressing these challenges requires interdisciplinary approaches spanning robotics, human-computer interaction, ethics, law, and policy.

The reliability of IoH systems is therefore paramount. Unlike IoT systems, where occasional data loss or delay may be tolerable, IoH requires end-to-end guarantees on latency, semantic consistency, and safety across perception, communication, and control loops. Meeting these requirements necessitates advances in embodied AI, deterministic

and 6G-enabled communication, semantic-level coordination, and resilient distributed system design tailored specifically to humanoid kinematics and human-centered interaction [1].

III. KEY ENABLING TECHNOLOGIES

This section outlines the core technologies essential to the development of IoH.

A. System Architecture

By decomposing humanoid systems into well-defined functional layers, IoH naturally admits a unifying architectural reference model that clarifies interfaces, responsibilities, and system-level guarantees. In this model, a humanoid system comprises five tightly coupled yet conceptually distinct layers: *perception*, *memory and world modeling*, *planning and decision*, *action*, and *networked coordination and cloud-edge*. This layered structure provides a common architectural vocabulary for integrating heterogeneous technologies while enabling modular development, verification, and evolution at scale.

- 1) *Perception Layer*: This layer aggregates local sensory inputs (e.g., vision, tactile, and proprioception) and networked observations acquired through IoH. Its responsibility is to produce time-aligned, uncertainty-aware state representations with explicit interfaces to downstream planning and learning modules. Standardized middleware [e.g., robot operating system 2 (ROS2)/data distribution service (DDS)] provides syntactic interoperability, while the layered reference model defines clear contracts for data freshness, provenance, and confidence, enabling perception outputs to be shared and reused across humanoids within the network.
- 2) *Memory and World Modeling Layer*: This layer maintains episodic memory, learned skills, and structured knowledge, supporting retrieval, abstraction, and reuse. Within the IoH reference model, this layer serves as the primary interface between local cognition and fleet-level intelligence, enabling shared world models, policy reuse, and experience aggregation. Responsibilities include versioning, consistency management, and knowledge provenance, which together provide guarantees for safe reuse across heterogeneous humanoids.
- 3) *Planning and Decision Layer*: This layer performs task reasoning, motion planning, and policy selection based on perceptual inputs and stored knowledge. Its interface contracts specify latency bounds, decision horizons, and fallback behaviors, ensuring that planning remains compatible with both real-time action constraints and network-level coordination. This layer enables local autonomy while supporting distributed task allocation and cooperative behaviors across the IoH.
- 4) *Action Layer*: This layer translates decisions into physical and social behaviors, including locomotion, manipulation, speech, and tool invocation. In the reference architecture, this layer provides strict real-time and safety guarantees, encapsulating embodiment-specific control loops behind standardized interfaces. This abstraction allows higher layers to coordinate actions

across humanoids without direct dependence on vendor-specific actuators or control implementations.

- 5) *Networked Coordination and Cloud-Edge Layer*: At the system level, IoH connects humanoids through cloud and edge infrastructure for coordination, learning, and governance. Rather than replacing local autonomy, this layer complements it by offering scalable services for task negotiation, shared learning, and global optimization. Emerging agent-level protocols and cloud-edge frameworks align naturally with the layered reference model by interfacing at well-defined boundaries (e.g., memory synchronization, task orchestration, or policy distribution).

Taken together, this layered decomposition constitutes a coherent architectural reference model for IoH. It enables seamless coupling of heterogeneous subsystems by clearly delineating responsibilities and interfaces across layers while providing explicit guarantees on timing, safety, and semantic consistency. This unified view synthesizes the surveyed technologies into an integrated system architecture, positioning IoH as a principled extension of robotics, networking, and AI rather than a loose collection of components.

1) *Artificial Intelligence*: AI is a foundational enabler of IoH because it allows humanoid robots to perceive, reason, learn, and act autonomously in complex, dynamic, and human-centric environments. As illustrated in Fig. 1, each humanoid robot functions as an intelligent node within the IoH ecosystem, capable of interacting with its physical surroundings while simultaneously collaborating with other humanoids through shared data, models, and networked decision-making.

Unlike traditional robotic systems that rely on preprogrammed behaviors or narrowly scoped perception pipelines, IoH humanoids increasingly leverage data-driven AI models that integrate perception, cognition, and action across multiple modalities. This shift enables higher levels of autonomy, adaptability, and collective intelligence, which are essential for large-scale deployment in unstructured real-world environments.

B. Large Multimodal Models

Large multimodal models (LMMs) have significantly expanded the capabilities of humanoid robots by enabling the joint processing and integration of diverse sensory inputs, including vision, speech, and touch. By learning shared representations across modalities, LMMs allow humanoids to ground language in perception and action, enabling more robust reasoning and context-aware interaction with humans and other robots [23].

Recent studies have demonstrated the effectiveness of LMMs in humanoid robotics. For example, Antikatzidis et al. [23] and Wang et al. [24] showed that LMM-based systems can process complex multimodal inputs in real time, enabling humanoids to navigate dynamic environments and engage in natural, intuitive human-robot interaction. Such models allow a humanoid to interpret a crowded scene, identify objects of interest, and respond to verbal or gestural cues (e.g., pointing or spoken commands). These capabilities are typically

achieved through large-scale multimodal pretraining followed by task- or embodiment-specific fine-tuning, which improves generalization across environments and tasks.

A notable advancement supporting LMM training is the Kitchen dataset, which provides a comprehensive multimodal benchmark for household manipulation tasks [25]. This dataset includes synchronized visual, tactile, auditory, and motion-capture data collected from human participants performing everyday cooking activities (e.g., cutting fruit and stirring ingredients). Training LMMs on such richly annotated datasets enables humanoids to learn long-horizon, multistep behaviors by imitating human demonstrations, thereby improving dexterity and robustness in domestic environments.

Wang et al. [26] explored the integration of large and multimodal language models—particularly GPT-4V—into robotic systems for task planning and reasoning. Their framework uses GPT-4V to generate step-by-step action plans from multimodal inputs (images and language), achieving strong performance across nine robotic datasets, including Google’s Open X-Embodiment. Reported matching scores (e.g., 9/10 for RT-1 and 10/10 for Freiburg Franka Play) demonstrate the model’s ability to understand instructions and environments. However, the study also highlights key limitations, such as reliance on predefined action sets, challenges in 3-D spatial reasoning, and sensitivity to prompt design, underscoring the gap between high-level reasoning and reliable physical execution.

Another line of work introduced *SplatTouch*, which tightly binds vision and tactile sensing to form explicit 3-D scene representations, improving robots’ ability to perform contact-rich and delicate manipulation tasks [27]. Such multimodal grounding is particularly important for humanoids, where small perception errors can lead to instability or unsafe physical interaction. In parallel, Google DeepMind’s Genie model represents a different but complementary advance [28]. Genie learns a latent action space from large-scale video data and can generate interactive environments from images, allowing robots to practice tasks in simulation before real-world deployment. By training on both game and robotics videos, Genie enables the learning of fine-grained control policies without explicit action annotations, supporting the development of generalist agents.

Within the IoH context, LMMs play a dual role: locally, which enhances individual humanoid autonomy and interaction quality; globally, which enables shared representations and knowledge transfer across fleets of humanoids. This makes LMMs a key technological bridge between embodied intelligence and networked collective learning.

The integration of LMMs into humanoid robotics has unlocked a wide range of applications, particularly in domains requiring sophisticated human-robot interaction. Table III summarizes the representative applications and the associated sensory modalities.

In caregiving scenarios, LMMs enable humanoids to interpret spoken requests and emotional cues, allowing them to respond appropriately—for example, fetching water while safely navigating obstacles. In service industries, LMM-equipped robots engage in natural language interaction and support tasks such as customer assistance and inventory

TABLE III
APPLICATIONS OF LMMs IN HUMANOID ROBOTICS

Application	Description	Modalities Used
Caregiving and Healthcare	Assist patients by interpreting verbal instructions, recognizing emotional cues, and performing tasks such as medication delivery.	Vision, Speech, Touch
Service and Hospitality	Interact with customers in hotels or retail settings, recognizing objects, gestures, and verbal commands for guidance and assistance.	Vision, Speech
Industrial Collaboration	Collaborate with human workers by understanding complex instructions and adapting to dynamic manufacturing environments.	Vision, Touch, Speech
Education and Training	Serve as interactive tutors, interpreting student queries and providing adaptive visual or verbal feedback.	Vision, Speech

handling. In industrial environments, LMMs facilitate close human–robot collaboration by enabling robots to understand abstract instructions and adjust to changing workspace conditions. In educational settings, LMM-powered humanoids can act as adaptive tutors, dynamically adjusting explanations and feedback based on student responses.

Despite their promise, the deployment of LMMs in humanoid robotics faces several fundamental challenges.

- 1) *Computational Complexity*: LMMs require substantial computational resources for both training and inference. This poses a critical challenge for real-time humanoid operation, where latency and power constraints limit the feasibility of deploying large models directly onboard. Optimizing model architectures, model compression, and leveraging edge–cloud offloading are therefore essential to achieving practical performance.
- 2) *Data Requirements*: Training LMMs demands large-scale, diverse datasets spanning multiple modalities. In humanoid robotics, collecting and annotating tactile, proprioceptive, and force-interaction data is particularly costly, limiting data availability compared to vision or language alone. This creates bottlenecks in scaling LMM training across different tasks and embodiments.
- 3) *Generalization Across Contexts*: Although LMMs perform well in controlled or benchmarked environments, generalization to unseen settings remains challenging. A humanoid trained in a structured laboratory environment may fail in cluttered, dynamic, or socially complex real-world spaces without additional adaptation, fine-tuning, or online learning mechanisms.
- 4) *Safety and Ethical Considerations*: Humanoids operate in close physical and social proximity to humans, making safety paramount. Misinterpretation of multimodal inputs or hallucinated reasoning by LMMs can translate directly into unsafe physical actions, especially in sensitive domains such as caregiving or healthcare. Ensuring predictable, verifiable, and ethically aligned behavior is therefore essential.
- 5) *Integration With Robotic Systems*: Translating high-level LMM outputs into precise, low-level motor commands

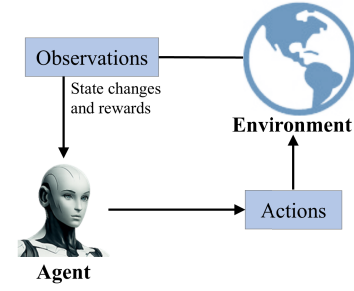


Fig. 3. RL.

remains nontrivial. Misalignment between symbolic or semantic reasoning and continuous control can lead to execution errors, instability, or degraded task performance, necessitating robust interfaces between perception, cognition, and control layers.

Looking forward, the evolution of LMMs in humanoid robotics—and in IoH systems more broadly—depends on addressing these challenges while enabling scalable, networked intelligence across humanoid fleets. Key research directions include the following.

- 1) *Efficient Model Architectures*: Designing lightweight and modular LMMs that preserve multimodal reasoning capabilities while reducing memory footprint and inference latency will be crucial for large-scale IoH deployment.
- 2) *Scalable Data Collection*: Advances in simulation, digital twins, synthetic data generation, and fleet-level data sharing can reduce reliance on expensive real-world data and enable collective learning across distributed humanoids.
- 3) *Hybrid Learning Approaches*: Combining LMMs with reinforcement learning (RL) and imitation learning (IL) enables humanoids to ground high-level reasoning in physical experience, improving adaptability while maintaining communication efficiency in fleet settings.
- 4) *Ethical and Safety Frameworks*: Developing principled frameworks for safety, robustness, and ethical alignment—integrated directly into LMM-driven perception and decision pipelines—is essential for building trust and ensuring responsible humanoid deployment.

Overall, LMMs provide a powerful foundation for embodied intelligence in IoH by enabling rich multimodal perception, reasoning, and interaction. Their successful integration into humanoid systems will require advances not only in model design but also in systems engineering, safety assurance, and network-aware learning. As these challenges are addressed, LMMs are poised to play a central role in enabling scalable, intelligent, and trustworthy humanoid ecosystems operating seamlessly within human-centric environments.

C. Reinforcement Learning

RL is a machine learning paradigm where an agent learns optimal actions by interacting with its environment, receiving rewards or penalties, and adjusting behaviors to maximize cumulative rewards, as shown in Fig. 3. In humanoid robotics,

RL is instrumental in optimizing motor skills requiring precise control and adaptability, such as walking, running, and object manipulation. For instance, Radosavovic et al. [29] demonstrated RL's effectiveness in controlling the locomotion of the humanoid robot Digit, achieving robust performance across diverse indoor and outdoor environments by training in simulation and deploying zero-shot in real-world settings. Similarly, Li et al. [30] advanced RL for versatile bipedal locomotion, addressing high-dimensional contact interactions to ensure stability in tasks like walking, running, and jumping on the Cassie robot.

While these works demonstrate strong single-robot performance, their direct applicability to *distributed fleet learning* in IoH is limited. In IoH settings, learning is often performed collaboratively across multiple humanoids with heterogeneous embodiments, sensors, and onboard compute capabilities, which motivates federated or decentralized training rather than centralized policy optimization.

Popular RL algorithms such as proximal policy optimization (PPO) [31] and soft actor critic (SAC) [32] are frequently employed. PPO, an on-policy algorithm, optimizes policies using a clipped surrogate objective, ensuring stable updates but requiring extensive samples for high-degree-of-freedom (DoF) systems. SAC, an off-policy algorithm, incorporates entropy regularization, promoting exploration but struggling with sparse rewards in tasks like simultaneous walking and manipulation. For example, Cheng et al. [33] used PPO to train expressive whole-body control for coordinated arm and leg movements in pushing tasks, while Dao et al. [34] applied sim-to-real RL for agile locomotion.

From a fleet learning perspective, both PPO and SAC impose significant communication burdens when extended to federated settings. PPO requires synchronized on-policy rollouts, making it poorly suited for asynchronous fleets. SAC, although off-policy, typically relies on sharing large replay buffers or dense gradient updates, which can overwhelm bandwidth-limited wireless networks common in IoH deployments.

In federated RL across humanoid fleets, communication of gradients or model parameters becomes a dominant bottleneck. Modern humanoid policies often contain millions of parameters, leading to tens or hundreds of megabytes per update. To mitigate this, gradient compression techniques—widely studied in federated learning—become essential. These include: 1) *quantization*, where gradients are transmitted at reduced precision (e.g., 8 or 1 bit), lowering bandwidth at the cost of biased updates; 2) *sparsification*, where only the top- k gradient components are communicated, reducing traffic but slowing convergence; and 3) *error-feedback mechanisms*, which accumulate dropped gradients locally to stabilize learning over time.

However, applying these techniques to humanoid RL introduces additional challenges. Humanoid data across a fleet are inherently non-IID due to differences in morphology, payloads, wear-and-tear, and operating environments. Naïve gradient averaging or aggressive compression can therefore destabilize learning or degrade performance on safety-critical control tasks. This highlights the need for morphology-aware

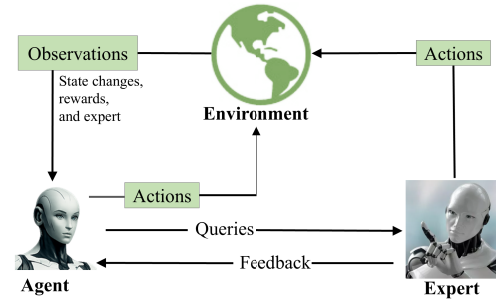


Fig. 4. IL.

aggregation, modular policy architectures, or shared latent representations that decouple common skills from robot-specific dynamics.

Nevertheless, RL often struggles in humanoid locomotion tasks due to task complexity, sample inefficiency, and sparse reward designs.

- 1) *High-Dimensional Action Spaces*: Humanoids with 20–40 DoF require policies to manage complex joint coordination, overwhelming PPO's on-policy updates and SAC's Q value estimations.
- 2) *Sim-to-Real Gap*: Training in simulation often fails to generalize due to discrepancies in contact models or sensor noise.
- 3) *Sample Inefficiency*: Both PPO and SAC demand extensive samples, impractical for physical humanoids where trials risk hardware damage [33].
- 4) *Sparse Rewards*: Tasks like door opening [35] provide sparse feedback, causing inefficient exploration.
- 5) *Safety Constraints*: RL exploration can lead to unsafe actions, requiring constrained optimization [36].
- 6) *Reward Engineering*: Designing rewards balancing locomotion stability and manipulation precision is challenging.

To address learning efficiency, approaches such as *offline RL* [37], *model-based RL* [38], and *curriculum learning* [39] are increasingly explored. In IoH fleet settings, offline and hybrid approaches are particularly attractive because they reduce communication frequency and allow policies to be updated using compressed or delayed gradients aggregated across robots. Integration with IL [40] further reduces exploration demands and improves stability.

Overall, while RL remains a powerful tool for humanoid control, its extension to distributed IoH fleet learning is fundamentally constrained by communication overhead, hardware heterogeneity, and safety considerations. These limitations motivate the growing emphasis on communication-efficient learning, gradient compression, and hybrid RL–IL strategies for scalable humanoid fleets.

D. Imitation Learning

IL empowers humanoid robots to acquire complex skills by observing and replicating human demonstrations [41], as shown in Fig. 4. It offers a rapid, intuitive, and efficient alternative to RL, which often relies on carefully engineered

reward functions. By leveraging human expertise, IL enables robots to learn intricate tasks—such as loco-manipulation, dexterous grasping, and visuotactile manipulation—that are difficult to specify via rewards. This approach is particularly valuable for applications requiring human-like dexterity, including caregiving, service roles, and industrial tasks.

From the perspective of distributed fleet learning in IoH, IL exhibits several advantages over RL. Most notably, IL can be trained using centralized or federated datasets of demonstrations without requiring frequent online interaction or synchronized rollouts across robots. This property significantly reduces communication overhead and makes IL inherently more amenable to large-scale fleet learning under bandwidth-constrained wireless networks.

In contrast to federated RL—where model gradients or parameters must be exchanged repeatedly—fleet-level IL often relies on aggregating demonstration data (e.g., trajectories, keyframes, or latent action representations) offline. When federated updates are required, IL models typically converge with fewer optimization rounds and lower gradient variance, enabling more aggressive gradient compression without destabilizing learning.

Specifically, gradient compression techniques such as low-bit quantization, top- k sparsification, and periodic model averaging are more effective for IL than RL in heterogeneous humanoid fleets. Because IL optimization is supervised and less sensitive to delayed or noisy updates, compressed gradients introduce less bias compared to value-based or policy-gradient RL methods. This robustness makes IL particularly suitable for federated learning across humanoids with differing DoF, sensors, and actuation characteristics.

The IL process typically involves collecting demonstration data through teleoperation, motion capture, or video analysis, which the robot then uses to infer action policies. Unlike RL, which optimizes a reward function, IL focuses on mimicking observed behaviors, making it particularly suited for tasks where human intuition and expertise are difficult to quantify.

For IoH-scale deployment, demonstration data can be collected asynchronously across fleets and aggregated centrally or via federated pipelines. Compared to RL rollouts, these data streams are less temporally coupled and more tolerant to delayed uploads, enabling practical deployment over intermittent or bandwidth-limited networks.

Recent research highlights the versatility of IL in enabling humanoid robots to perform complex tasks. Seo et al. [42] introduced the transparent interconnection of lots of links (TRILL) framework using VR teleoperation to collect high-quality demonstrations for loco-manipulation. He et al. [43] extended IL to tactile manipulation using behavior cloning. Although they demonstrated on quadrupeds, their multimodal learning insights generalize to humanoids operating in contact-rich environments.

The effectiveness of IL hinges on the quality and diversity of demonstration data. Common data sources include teleoperation, motion capture, and human videos. In fleet learning settings, combining heterogeneous demonstrations from multiple humanoids further amplifies coverage of task variations, improving generalization without increasing communication frequency.

Despite its promise, IL faces challenges.

- 1) *Data Quality and Quantity*: Demonstration collection remains labor-intensive.
- 2) *Generalization*: Policies may fail in unseen scenarios without hierarchical or adversarial training.
- 3) *Safety and Robustness*: Edge cases require explicit safety-aware imitation.

Overall, IL is particularly well-suited for distributed fleet learning in IoH due to its communication efficiency, tolerance to compressed and asynchronous updates, and robustness under heterogeneous robot embodiments. These properties make IL a natural backbone for scalable humanoid fleet learning, either as a standalone approach or as the foundation for hybrid IL–RL systems discussed next.

E. Hybrid Approaches Combining RL and IL

Hybrid approaches combining RL and IL leverage the complementary strengths of both paradigms: fast skill acquisition from demonstrations and robustness through trial-and-error refinement. In these approaches, IL provides an initial policy (from human or planner demonstrations), and RL subsequently refines this policy through environmental feedback, enabling adaptation and generalization to novel scenarios. For example, Zhang et al. [46] developed a plan-guided RL framework for whole-body manipulation on a humanoid, where a motion plan initializes the policy and PPO-based RL refines it, yielding robust performance in contact-rich tasks. Related teacher–student schemes have proven effective for sim-to-real transfer; an IL-trained policy is first learned from expert data and then fine-tuned with RL to exceed the original demonstrations and adapt to new conditions [47]. These hybrid methods excel at tasks demanding precision and adaptability, such as industrial assembly and caregiving involving deformable or unpredictable objects [47], [48], [49].

From the perspective of distributed fleet learning in IoH, hybrid IL+RL approaches offer a particularly attractive trade-off between learning efficiency and communication cost. Specifically, the IL phase can be trained centrally or via federated learning with aggressive gradient compression, while the subsequent RL fine-tuning can be performed either locally on individual humanoids or with sparse, low-frequency federated updates. This separation substantially reduces bandwidth requirements compared to fully federated RL pipelines.

In heterogeneous humanoid fleets—where robots differ in degrees of freedom, sensor suites, and actuation dynamics—pure federated RL is often unstable under gradient compression due to non-IID data and high gradient variance. Hybrid approaches mitigate this issue by anchoring learning to a shared IL-initialized representation, which stabilizes subsequent RL updates and makes compressed or partially synchronized gradients more effective. As a result, techniques such as top- k sparsification, low-bit quantization, and periodic model averaging are more viable in hybrid settings than in standalone RL.

Moreover, hybrid learning naturally supports hierarchical federated architectures; low-level motor primitives learned via IL can be shared fleet-wide with minimal communication, while higher level RL policies adapt locally to embodiment-

TABLE IV
RL, IL, AND HYBRID LEARNING APPROACHES

Learning	Key Tasks	Representative Works	Training Scale	Reported Performance	Fleet Learning Implications
RL	Locomotion, whole-body manipulation	Radosavovic <i>et al.</i> (Digit locomotion) [29]; Li <i>et al.</i> (Cassie) [30]; Cheng <i>et al.</i> [33]; DDF [34]	10^7 – 10^9 simulation steps; days to weeks of GPU training	90–98% success in simulation; 70–85% zero-shot sim-to-real transfer; noticeable degradation under morphology or environment shift	High gradient variance; large model updates (>10–100 MB/round); sensitive to quantization and sparsification; non-IID data across heterogeneous humanoids destabilizes federated RL
IL	Loco-manipulation, grasping, visuotactile manipulation	TRILL [42]; He <i>et al.</i> [43]; Yagi <i>et al.</i> [44]; Zhao <i>et al.</i> [45]	10^2 – 10^4 demonstrations; hours to days of training	80–95% success on demonstrated tasks; sim-to-real degradation typically <10–15% when physical demonstrations are used	Highly communication-efficient; supervised objectives tolerate aggressive gradient compression (top- k , 8-bit); well suited for asynchronous federated fleet learning
Hybrid (IL+RL)	Whole-body manipulation, sim-to-real transfer	Plan-guided RL [46]; Warm-start RL [47]; Adaptive hybrid learning [48]	IL pretraining (10^2 – 10^3 demos) + RL fine-tuning (10^6 – 10^8 steps)	85–95% real-world task success; 20–40% reduction in sim-to-real gap compared to pure RL	Most favorable for IoH fleets: IL stabilizes representations, enabling compressed and low-frequency federated updates with local RL refinement

specific dynamics or task contexts. This hierarchy is well aligned with IoH deployments, where global knowledge sharing and local autonomy must coexist under wireless bandwidth and latency constraints.

Despite significant advancements, applying RL and IL—either individually or in hybrid form—to humanoid robotics still faces challenges. One major issue is the sim-to-real gap: humanoids’ high-dimensional, contact-rich dynamics make it nearly impossible to perfectly match simulation to reality, requiring extensive domain randomization and tuning [21], [50]. Consequently, RL training remains data hungry, with sparse and complex reward signals leading to high sample complexity [51]. Unstable nonlinear contact dynamics further complicate learning and can violate safety limits during exploration [52].

While hybrid IL+RL alleviates some of these issues, federated deployment at IoH scale remains constrained by demonstration collection costs, residual sim-to-real mismatch, and the need to balance local adaptation against globally shared models. These limitations highlight the importance of communication-efficient hybrid learning pipelines—combining IL initialization, compressed federated updates, and selective local RL refinement—as a practical path toward scalable humanoid fleet learning.

Table IV summarizes RL, IL, and hybrid learning techniques for humanoid robotics.

1) *Sensors and Actuators*: Humanoid robots rely on a highly integrated suite of sensors and actuators to perceive, interpret, and act upon their environments. Because humanoids operate as embodied, force-exerting agents in human-centric spaces, sensing and actuation are tightly coupled with safety, autonomy, and network-level coordination in IoH. Progress in this domain has led to remarkable improvements in dexterity, balance, and robustness. Table V provides a technical overview of state-of-the-art sensor technologies (tactile, proprioceptive, visual, and inertial) and actuation systems (electric, compliant, and soft), with emphasis on scalability, latency, and reliability requirements unique to humanoid fleets. Emerging trends such

as neuromorphic sensing and electroactive actuation are also discussed.

F. Tactile Sensing and Artificial Skin

Tactile sensing is crucial for achieving human-like physical interaction, enabling humanoids to detect contact forces, shear, texture, and temperature during manipulation and locomotion. At the IoH scale, tactile systems must further support distributed sensing, low-latency feedback, and robustness to long-term wear.

Capacitive tactile sensors offer high sensitivity and robustness to electrical noise, making them well-suited for high-resolution artificial skin on curved humanoid surfaces [53]. In contrast, piezoresistive skins are simpler to fabricate and integrate with flexible substrates, although they exhibit higher hysteresis and temperature dependence [54].

Bao *et al.* [55] proposed the integrated intelligent tactile system (IITS), which features a multichannel electronic skin with real-time feedback and closed-loop control for dexterous manipulation. Another promising direction is neuromorphic tactile sensing, which encodes contact events as asynchronous spikes rather than dense signals. For example, Lee *et al.* [56] developed a spiking tactile neuron array with submillisecond latency, enabling fast reflexive responses in humanoids. Such approaches are particularly attractive for IoH deployments where low-latency local reactions must coexist with network-level coordination.

G. Force/Torque Sensors and Calibration Advances

Multiaxis force/torque (F/T) sensors are essential for whole-body control, impedance modulation, and safe human–robot interaction. Recent trends emphasize miniaturization, high bandwidth, and improved calibration to support high-DoF humanoid manipulation and locomotion.

Mao *et al.* [57] presented an ultralight (0.30 g) fingertip-scale six-axis F/T sensor based on thermistor networks, enabling omnidirectional force sensing at the fingertips. Mohamed and Singh [58] addressed calibration challenges

by introducing a nonlinear learning-based model that reduces root-mean-square error (RMSE) by approximately 50% compared with affine baselines. Soft, strain-based force sensors are also being explored for compliant humanoid joints, offering improved safety and adaptability at the cost of increased modeling complexity [59].

H. Inertial and Proprioceptive Sensing

Inertial measurement units (IMUs), joint encoders, and proprioceptive sensors form the backbone of humanoid balance, state estimation, and locomotion control. However, MEMS-based IMUs are prone to drift, vibration, and shock-induced errors.

Allione et al. [70] demonstrated that even low-impact footfalls can significantly distort heading estimates during humanoid walking. To mitigate this, Yang et al. [60] employed multiple IMUs distributed across robot links to correct dominant error sources in standard proprioceptive odometry, achieving substantial accuracy improvements. Complementing these approaches, distributed proprioceptive skins embedded within joints provide continuous, spatially resolved feedback along limbs, improving robustness for whole-body control in dynamic environments [61].

I. Vision and Exteroceptive Sensing

Visual and exteroceptive sensing has evolved from conventional RGB and RGB-D cameras toward high-temporal-resolution, event-driven, and multimodal perception systems that enable faster reaction times, lower latency, and improved energy efficiency—properties that are particularly critical for humanoid locomotion and manipulation in dynamic environments.

D'Angelo et al. [62] integrated event-based cameras into the iCub humanoid control stack, achieving dynamic scene segmentation at rates exceeding 500 Hz while significantly reducing computational load compared to frame-based vision. Such neuromorphic vision systems are well-suited for reflexive behaviors and low-latency control loops, complementing slower but semantically richer perception pipelines. In parallel, fusion of depth sensing with semantic segmentation has enabled anticipatory motion planning in cluttered and human-populated environments, allowing humanoids to reason about obstacles, affordances, and free space before physical contact occurs [63]. These hybrid perception strategies highlight a broader trend toward layered visual sensing, where fast reactive perception and high-level semantic understanding coexist within a unified architecture.

J. Electric Actuators and Transmission Systems

Most state-of-the-art humanoid robots rely on high-performance brushless dc (BLDC) or permanent magnet synchronous motors coupled with compact transmission systems such as planetary gearboxes or harmonic drives. These actuation solutions are favored for their high power density, controllability, and compatibility with electrically driven humanoid platforms.

Recent designs increasingly emphasize torque density and mass efficiency at the joint level. For example, Mithra's humanoid incorporates a compact BLDC actuator combined with a four-bar transmission mechanism, enabling high output torque at reduced weight and inertia [64]. Harmonic drives provide high reduction ratios in compact form factors but are susceptible to backlash, elastic deformation, and long-term fatigue. As an alternative, series elastic actuators (SEAs) introduce compliant elements between the motor and load, improving shock tolerance, force control accuracy, and safety during physical human–robot interaction [65]. These tradeoffs between stiffness, efficiency, and robustness directly influence the feasibility of whole-body control and safe deployment in IoH scenarios.

K. Soft and Variable-Stiffness Actuators

Compliant and soft actuation is gaining increasing importance in humanoid robotics due to its potential benefits in energy efficiency, adaptability, and intrinsic safety. Unlike rigid electric actuators, soft and variable-stiffness actuators more closely emulate biological musculoskeletal systems, enabling smoother interaction with uncertain environments and humans.

Emerging approaches include cable-driven artificial muscles, dielectric elastomer actuators (DEAs), and magnetically actuated soft mechanisms. Seong et al. [66] introduced a bioinspired magnetic actuator whose stiffness can be dynamically modulated via external magnetic fields, enabling rapid transitions between compliant and rigid behaviors. Similarly, variable-stiffness joints based on shape-memory alloys or jamming transitions allow humanoid joints to adapt their mechanical impedance to task demands, balancing precision and safety [67].

While these technologies remain less mature than conventional electric actuation, their integration into future humanoid platforms is expected to play a key role in achieving resilient, energy-aware, and human-compatible motion—particularly in IoH deployments where long-term autonomy and safety are paramount.

L. Emerging Trends

Emerging technologies increasingly aim to tighten the coupling between sensing, computation, and actuation, giving rise to artificial neuromuscular systems inspired by biological reflexes. For example, neuromorphic chips that integrate tactile and proprioceptive input layers are being embedded directly into humanoid reflex arcs, enabling submillisecond response times and adaptive, event-driven control [69]. By processing sensory signals asynchronously and locally, such architectures reduce latency and reliance on centralized computation, which is critical for stability and safety in dynamic humanoid motion.

In parallel, electroactive polymers (EAPs) are maturing as lightweight, low-voltage actuators capable of silent, muscle-like contraction and continuous deformation [68]. These actuators offer intrinsic compliance and high power-to-weight potential, making them attractive for next-generation humanoid joints, wearable exoskeletons, and soft interaction

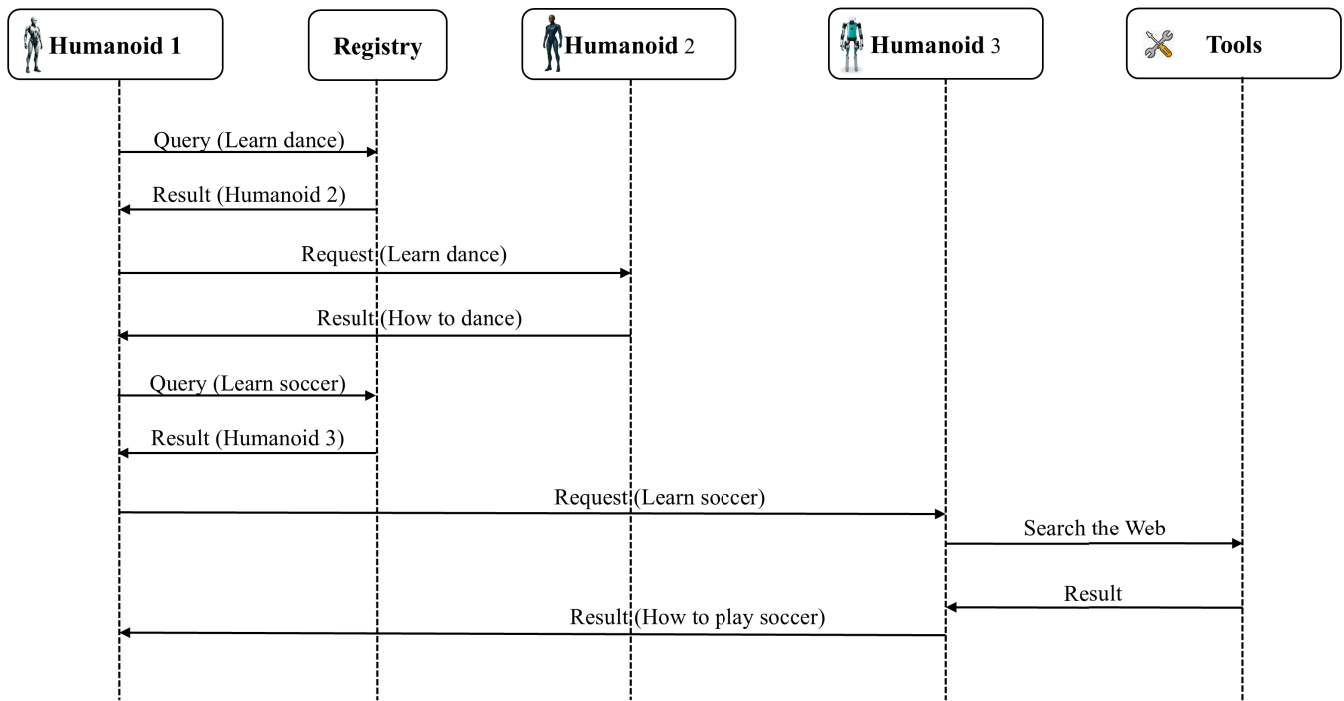


Fig. 5. Message sequence diagram of the IoH.

interfaces, despite current limitations in durability and control precision.

Collectively, these trends point toward a shift from rigid, centrally controlled architectures to distributed, embodied sensing–actuation loops, where intelligence is partially embedded at the hardware level. Such designs are expected to play a central role in the post-2025 humanoid roadmap, particularly for IoH deployments that demand low latency, energy efficiency, resilience, and scalable coordination across large fleets of embodied agents.

1) *Communication Protocols*: IoH requires sophisticated communication protocols to facilitate efficient, secure, and interoperable exchanges among humanoid robots, distributed AI agents, external tools, and human users operating in complex, unstructured, and dynamic environments.

M. Traditional Protocols

IoH can build upon well-established protocols from the IoT domain, such as MQTT and CoAP while exploiting advanced network technologies like 5G and emerging 6G standards to meet the demands of high-bandwidth, low-latency applications. The lightweight publish–subscribe architecture of MQTT supports real-time streaming of proprioceptive and exteroceptive sensor data across multiple agents and devices, enabling distributed perception and decision-making. Meanwhile, 6G networks promise ultrareliable communication with latencies below 1 ms, making them ideal for high-speed teleoperation, swarm coordination, and safety-critical behaviors in humanoid collectives [71]. Complementing these protocols, edge computing plays a critical role by offloading computation from the cloud to local processing nodes embedded near or within the humanoid robot. This approach significantly reduces

the round-trip delay associated with cloud communication and ensures the timely execution of control loops. For example, in scenarios where a humanoid must avoid a fast-approaching obstacle or respond to a sudden physical interaction with a human, edge-enabled systems ensure sub-10-ms response times, which are essential for safe, reactive behaviors [72]. The synergy between real-time networking, edge intelligence, and protocol-level interoperability forms the backbone of scalable, responsive, and secure IoH ecosystems.

Nevertheless, unlike traditional networked systems, humanoid robots must orchestrate multimodal sensing, actuation, and high-level decision-making in real time—often spanning heterogeneous hardware platforms, simulation environments, and organizational boundaries. These interactions require deterministic, low-latency communication and interoperability at both the syntactic and semantic levels, accommodating diverse data types such as multicamera RGB-D video, haptic feedback, joint torque profiles, and high-level symbolic commands. However, existing IoT protocols like MQTT and CoAP, optimized for low-bandwidth, stateless, and device-to-device communication, were primarily designed for predictable telemetry and simple control flows. In contrast, IoH involves humanoid robots powered by large models, which exhibit complex, context-dependent behaviors such as task decomposition, multiturn dialogs, and adaptive decision-making. These advanced capabilities demand richer, peer-to-peer, stateful, and context-aware interactions that IoT protocols alone cannot provide. Fig. 5 illustrates a representative message sequence, showing a humanoid robot engaging in multiagent communication—coordinating with peer humanoids and interacting with external digital systems (e.g., search engines)—to complete a collaborative task in a dynamic environment.

Recent advances in agent-to-agent (A2A) communication protocols, introduced by Google, and the model context protocol (MCP), developed by Anthropic, offer promising solutions to address these needs [73].

N. A2A Communication Protocol

Google's A2A protocol introduces a standardized HTTP/JSON-remote procedure call (RPC) communication framework enriched by agent cards—structured metadata describing an agent's identity, capabilities, security posture, supported tasks, and artifact formats. Agents can discover compatible peers, initiate long-running workflows, handle asynchronous outcomes, and exchange provenance-rich artifacts with enterprise-grade authentication baked in. A2A is designed for horizontal scalability: agents across different platforms, vendors, or even digital twins can discover and negotiate subtasks transparently, without bespoke adapters or custom application programming interfaces (APIs), simplifying cross-domain orchestration and monitoring across fleets. Its adoption by the Linux Foundation (LF) further reinforces its neutrality and broad governance, allowing collective evolution of the standard beyond a single vendor's roadmap [74], [75], [76].

From a performance perspective, A2A is intentionally optimized for semantic coordination rather than real-time control. A2A relies on text-based JSON serialization over HTTP or similar transports, which introduces nontrivial latency and bandwidth overhead compared to binary serialization mechanisms (e.g., DDS-CDR, Protobuf, and FlatBuffers) commonly used in robotic control middleware. In high-DoF humanoid systems, joint-state vectors alone may exceed hundreds of floating-point values per cycle; serializing such data as JSON increases payload size by 3–10 $\times$ and incurs additional parsing overhead, making A2A unsuitable for millisecond-level control loops.

However, this limitation is architectural rather than accidental. A2A is explicitly designed for task-level negotiation, agent discovery, workflow orchestration, and provenance tracking, where latencies on the order of tens to hundreds of milliseconds are acceptable. In this regime, the benefits of human-readable schemas, introspection, security, and cross-vendor interoperability outweigh raw transport efficiency.

O. Model Context Protocol

Complementing A2A, Anthropic's MCP defines a structured JSON-RPC interface that wraps external tools, APIs, and resources behind a consistent endpoint, complete with authentication, usage logging, failure semantics, and structured output schemas. Dubbed the “USB-C port for AI,” MCP abstracts away the heterogeneity of backend systems—databases, search engines, robotic tools, and geographic APIs—so that agents need only interact with a common protocol surface. This dramatically reduces integration cost, improves security through least-privilege access, and enforces uniform logging and auditing policies. Because every MCP-exposed capability adheres to the same RPC-based pattern, agents can introspect, plan, and reason over tool capabilities in a generalized, scalable way [77], [78], [79].

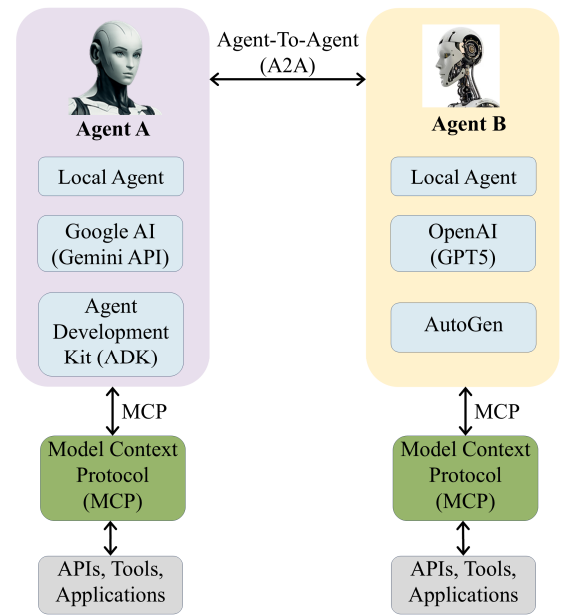


Fig. 6. Humanoid communications with A2A and MCP.

Similar to A2A, MCP is not intended for streaming high-frequency control signals or sensor feedback. The use of JSON-based request–response semantics introduces overhead that is incompatible with real-time humanoid control loops requiring submillisecond latency and deterministic jitter bounds. Binary middleware used in robotics (e.g., DDS/RTSPS with zero-copy transport) achieves orders-of-magnitude lower serialization cost and predictable timing guarantees, which MCP deliberately does not target.

Instead, MCP occupies a higher abstraction layer, where calls represent discrete semantic actions (e.g., “plan grasp,” “query map,” and “invoke perception module”) rather than continuous control trajectories. In this context, bandwidth efficiency is dominated by semantic payloads (text, symbolic structures, or compressed artifacts), and the dominant performance requirement becomes robustness, traceability, and fault isolation rather than raw throughput.

P. IoH With A2A and MCP

For IoH—where autonomous humanoid robots collaborate, interact with humans, and leverage dynamic tool ecosystems—the A2A + MCP duo offers a potent architectural pattern, provided that it is correctly positioned outside safety-critical real-time control loops. Fig. 6 illustrates humanoid communications with A2A and MCP.

- 1) *Tool Encapsulation With MCP*: Wrap each humanoid robot's internal capabilities—motion planners, grasp synthesis modules, visual–semantic query engines, and local mapping services—behind MCP endpoints. These endpoints operate at the level of planning, perception, or symbolic reasoning, while time-critical joint control and whole-body stabilization remain local and binary-serialized within the robot. This allows external agents or orchestrators to issue safe, auditable tool invocations, retrieve structured responses, and gracefully handle failures or resource constraints.

- 2) *Agent Discovery and Coordination via A2A*: Use A2A's discovery mechanisms to locate relevant humanoid agents; each humanoid robot publishes an agent card outlining its current state, tool-wrapping MCP endpoints, geographic reach, and workload. Orchestrator or peer agents can then negotiate task allocations ("deliver package to station B" and "assist human operator with assembly") using A2A's task-lifecycle protocols. Because these negotiations occur on human and mission timescales, the added latency of text-based protocols does not interfere with physical stability or safety.
- 3) *Separation of Concerns—Context Access Versus Coordination*: MCP handles what an agent can do and exposes tools securely; A2A handles who coordinates those capabilities and how tasks are negotiated and tracked. This separation mirrors a control-plane/data-plane split; A2A and MCP form the semantic control plane, while real-time binary middleware (e.g., ROS2/DDS) remains the data plane for high-rate sensor and actuator streams.
- 4) *Provenance, Observability, and Governance*: With MCP logging each tool call and A2A tracking task initiation, artifact flow, and agent attribution, operators gain end-to-end observability—from mission-level goals to tool usage. Such observability would be prohibitively expensive to implement inside real-time binary control loops, further justifying the use of higher level text-based protocols at this layer.
- 5) *Scalability and Cross-Vendor Federation*: As IoH ecosystems scale across vendors and operators, standardized MCP and A2A allow interoperability regardless of robot origin. Crucially, this scalability is achieved without burdening real-time control paths with human-readable protocols, preserving safety and performance while enabling semantic-level federation.

In summary, A2A and MCP should not be viewed as replacements for real-time robotic middleware, but as complementary semantic-layer protocols. Binary serialization and deterministic transport remain indispensable for high-DoF control loops, whereas A2A and MCP enable scalable coordination, tool abstraction, governance, and interoperability at the system level. Together, this layered architecture supports a composable, vendor-neutral, and observability-first IoH infrastructure that respects the fundamental performance constraints of humanoid robotics [74], [75], [78], [79].

1) *Cloud and Distributed Systems*: Cloud and distributed infrastructures are central to realizing the IoH, providing the compute, storage, coordination, and learning substrates that allow humanoids to operate as collaborative, knowledge-sharing agents rather than isolated devices. Mature cloud platforms and open toolchains—such as AWS RoboMaker for large-scale simulation and fleet workflows (end-of-support September 10, 2025) and Google's open-source Cloud Robotics Core built on Kubernetes—enable offloading of latency-tolerant and resource-intensive workloads, including high-dimensional motion planning, large-model inference, global mapping, and dataset distillation. In contrast, reflexive and safety-critical control loops remain on-robot or at the edge

[80], [81], [82]. This split-compute paradigm is a defining architectural principle of IoH, matching task requirements to robot, edge, and cloud tiers to jointly optimize latency, energy efficiency, and scalability.

Beyond raw compute, distributed streaming systems and robotics middleware are essential for orchestrating multimodal data flows and interhumanoid coordination. Widely adopted data infrastructures such as Apache Kafka (high-throughput pub/sub logs) and Apache Flink (stateful stream processing) provide durable backbones for aggregating proprioceptive, visual, and haptic streams, supporting event-driven pipelines for teleoperation, fleet monitoring, and large-scale policy roll-outs [83], [84]. Within the robotics stack, ROS2 transports rely on the DDS to deliver reliable, real-time QoS semantics across heterogeneous nodes. Recent studies highlight how DDS-level QoS tuning (latency, reliability, and deadline policies) is critical for maintaining closed-loop stability and interoperability in large humanoid fleets [85]. Together, these layers—streaming backbones, stream processors, and DDS-based robotics middleware—form the data plane and control plane of scalable IoH sensing–actuation loops.

To achieve elasticity and fault tolerance at fleet scale, IoH systems increasingly adopt microservices and container orchestration frameworks such as Kubernetes. Recent work demonstrates ROS2 services packaged as containerized microservices that can be autoscaled, migrated, and recovered under node or network failures [86]. This design enables graceful degradation and rapid failover, ensuring that partial subsystem failures (e.g., simultaneous localization and mapping (SLAM), perception, or planning services) do not cascade into full mission failure—an essential requirement for safety-critical humanoid deployments.

Edge and fog computing now constitute a standard layer in humanoid system design, yielding hybrid robot–edge–cloud pipelines. Latency-critical loops—such as balance control, contact-rich manipulation, and safety monitoring—execute at edge sites proximal to the robot, while compute-intensive training and aggregation tasks (e.g., multihumanoid policy learning and cross-domain knowledge distillation) are performed in the cloud. Frameworks such as FogROS2 and FogROS2-SGC demonstrate how ROS2 workloads can be dynamically placed across edge and cloud resources with adaptive scheduling to reduce jitter and end-to-end latency [87], [88]. This hierarchical deployment aligns with tactile-internet objectives—ultralow latency and high reliability—which are particularly relevant for humanoid haptics, force feedback, and telemanipulation [89], [90].

Cloud backends are increasingly converging with VLA and foundation models for instruction following, semantic scene understanding, and multiturn planning. Recent systems such as RT-2 illustrate how web-scale pretraining combined with robotic trajectories can yield generalizable control policies, but only when supported by scalable cloud inference, semantic middleware, and structured interfaces bridging low-level signals (torques and tactile feedback) and high-level abstractions (goals, affordances, and task graphs) [22]. Finally, collaborative learning is naturally deployed over these distributed substrates. Communication-efficient federated RL enables

TABLE V
COMPARISON OF SENSOR AND ACTUATOR TECHNOLOGIES FOR HUMANOID ROBOTICS

Component Type	Subcategory	Technical Description (with References)	Key Advantages and Challenges
Tactile Sensing	Capacitive	High-resolution electronic skin with low noise and curvature compatibility [53].	High sensitivity; vulnerable to electromagnetic interference.
	Piezoresistive	Flexible, low-cost skins with scalable fabrication [54].	Simple integration; hysteresis and thermal drift.
	Neuromorphic	Spiking tactile arrays with sub-ms latency [56].	Ultra-low latency; complex neuromorphic hardware.
Force/Torque Sensing	Fingertip F/T	Miniaturized six-axis thermistor-based sensor (0.3 g) [57].	Lightweight and compact; nonlinear response.
	Calibration Methods	Learning-based calibration achieving $\sim 50\%$ RMSE reduction [58].	Improved accuracy; data-intensive training.
Inertial & Proprioceptive	Sensor Fusion	Joint encoders and IMU fusion reducing drift by 79% [60].	High robustness; increased computational cost.
	Joint Skins	Distributed proprioception via embedded sensing skins [61].	Enhanced feedback; complex mechanical integration.
Visual Sensing	Event-Based Cameras	Event-driven perception exceeding 500 Hz [62].	Ultra-fast response; algorithmic complexity.
	Vision Fusion	Semantic-depth fusion for cluttered navigation [63].	Rich perception; GPU dependency.
Electric Actuators	BLDC Motors	Compact BLDC actuator with four-bar transmission for high torque density [64].	High efficiency; requires precise control.
	Harmonic Drive	High-ratio compact transmission [65].	Precision; fatigue and backlash risks.
Compliant Actuators	Series Elastic	Modular SEAs for safe interaction [65].	Shock tolerance; added mass.
	Magnetic Actuators	Field-modulated stiffness in bioinspired designs [66].	Adaptive compliance; limited power density.
Soft Actuators	Variable-Stiffness Joints	SMA-based adaptive stiffness joints [67].	Muscle-like behavior; slow thermal response.
	Electroactive Polymers	Low-voltage soft artificial muscles [68].	Lightweight and silent; durability challenges.
Emerging Technologies	Neuromorphic Reflex Chips	Tactile-proprioceptive neurosystems enabling low-latency reflexes [69].	Fast local reactions; early-stage integration.

TABLE VI
CLOUD AND DISTRIBUTED COMPUTING INFRASTRUCTURES ENABLING THE IOH

Layer / Component	Role in IoH	Representative Frameworks / Examples	References
Cloud robotics platforms	Large-scale simulation, fleet management, global mapping, and foundation-model inference/training	AWS RoboMaker (EoS 2025), Google Cloud Robotics Core (Kubernetes-based)	[80]–[82]
Data streaming backbones	High-throughput pub/sub logs for multimodal robot data; event-driven fleet analytics	Apache Kafka, Apache Flink	[83], [84]
Robotics middleware (ROS2 + DDS)	Real-time QoS semantics, heterogeneous inter-robot communication, interoperability	Fast DDS, Cyclone DDS, RTI Connext	[85]
Microservices & orchestration	Elasticity, fault tolerance, autoscaling of robotic services	Kubernetes-based ROS2 microservices	[86]
Edge / fog computing	Ultra-low-latency control at edge; cloud for training and global optimization	FogROS2, FogROS2-SGC	[87], [88]
Tactile-internet alignment	Sub-ms latency and high reliability for haptics and teleoperation	Edge-assisted tactile loops	[89], [90]
Cloud–foundation model integration	Scaling VLA/VLM inference for language-conditioned planning and perception	RT-2 and related VLA systems	[22]
Collaborative learning substrates	Federated RL and decentralized imitation learning without raw data exchange	Communication-efficient federated RL	[91]

policy sharing across geographically dispersed humanoids without exchanging raw sensory data, mitigating both bandwidth and privacy constraints [91]. Together, cloud, edge, streaming, and learning infrastructures form the backbone of scalable, resilient, and adaptive IoH ecosystems.

Table VI summarizes the cloud and distributed computing infrastructures enabling IoH.

IV. APPLICATIONS

IoH offers transformative potential across a wide range of real-world domains by enabling humanoid robots to collaborate, adapt, and operate as intelligent partners rather

than isolated machines. By interconnecting perception, learning, and decision-making across fleets of humanoids, IoH enables collective intelligence, experience sharing, and coordinated task execution in dynamic human-centric environments. This represents a paradigm shift from standalone robotic deployments to networked humanoid ecosystems capable of continuous improvement and large-scale coordination. Table VII summarizes representative IoH applications across diverse domains.

A. Elder Care and Assisted Living

Humanoid robots are increasingly explored as comprehensive care partners for older adults, providing emotional

TABLE VII
APPLICATIONS OF THE IOH ACROSS DIVERSE DOMAINS

Application Area	Role in IoH	Representative Systems / Examples	References
Elder care & assisted living	Emotional companionship, mobility assistance, medication reminders, cognitive training; personalized care via shared models	Ka Ka household companion robot; tailored humanoid assistants	[92]–[94]
Healthcare & teleoperated medical assistance	Remote medical procedures, hospital logistics automation, coordinated patient care	Unitree G1 for tele-surgery; Diligent Robotics’ Moxi in U.S. hospitals	[95], [96]
Education & special needs support	Multi-modal robotic teaching assistants; adaptive curricula and pedagogy sharing	Pepper robot in classrooms; gesture-based tutoring systems	[97], [98]
Social companionship & emotional support	Stress reduction, loneliness mitigation, therapeutic engagement	PARO; humanoid social robots with self-disclosure protocols	[100]
Household assistance & domestic robots	Cleaning, serving, and manipulation tasks; shared maps, skills, and preferences across homes	German SMILE project (Justin, EDAN); Honda T-HR3; Care-O-bot	[101]
Research & embodied cognition	Studying embodied intelligence, dexterity, and learning from interaction; open research platforms	iCub; Salvius open-source humanoid	[101], [102]
Industry environments (smart factories)	Adaptive humanoid co-workers integrated with IIoT; workflow optimization	Humanoids for assembly, logistics, inspection	[103]
Public engagement & competitions	Benchmarking agility, real-world task performance, and public awareness	2025 World Humanoid Robot Games (Beijing)	[104]

companionship, medication reminders, mobility assistance, and cognitive stimulation. A comprehensive scoping review by Andtfolk et al. [92] examined humanoid deployments in elder care settings and found them particularly effective in supporting daily activities, facilitating social interaction, and enabling cognitive training exercises. Within an IoH context, such capabilities can be enhanced by sharing personalized care strategies, behavioral models, and anonymized usage data across multiple deployments, enabling continual refinement of assistive behaviors.

However, the review also highlighted persistent challenges, including slow response times, technical malfunctions, and declining user engagement overextended periods. These limitations motivate IoH-level solutions, such as cloud-assisted perception, fleet-level monitoring, and cross-robot learning to improve robustness and responsiveness over time.

Building on these findings, qualitative studies indicate that older adults and physically disabled users particularly value humanoid assistants when systems are carefully tailored to individual preferences and routines [93]. IoH architectures naturally support such personalization by allowing models and care policies to be adapted locally while benefiting from shared learning across similar user profiles.

Longitudinal studies of household companion robots, such as the “Ka Ka” system, demonstrate measurable reductions in loneliness and improved social engagement during extended home deployments [94]. These results suggest that long-term, networked humanoid companionship—supported by IoH-level updates, diagnostics, and behavioral sharing—can yield sustained psychological benefits, provided that technical reliability and privacy are carefully managed.

B. Healthcare and Teleoperated Medical Assistance

The healthcare sector represents one of the most promising and high-impact frontiers for IoH deployment, as it combines stringent safety requirements, real-time interaction, and strong

demand for scalable assistance. Recent advances in teleoperated humanoid systems illustrate how IoH can extend medical expertise beyond physical boundaries while preserving embodied interaction. A landmark 2025 technical study demonstrated direct bimanual teleoperation of a Unitree G1 humanoid robot performing complex medical procedures, including patient palpation, ventilation support, and ultrasound-guided injections [95]. While this work highlights the feasibility of remote, embodied medical intervention, it also exposes critical limitations in haptic fidelity, motion precision, latency control, and network reliability that must be resolved before clinical-grade deployment.

In parallel, semi-autonomous humanoid service robots have begun transitioning from experimental prototypes to operational healthcare infrastructure. For example, Diligent Robotics’ Moxi is currently deployed in hospitals across USA, where it autonomously delivers medical supplies, transports laboratory samples, and performs routine logistics tasks [96]. Although Moxi operates primarily as a task-focused service robot, its success demonstrates the near-term value of humanoid platforms integrated into hospital workflows, particularly when coordinated with digital hospital information systems.

IoH introduces a qualitatively new capability for healthcare systems: coordinated, multihumanoid medical assistance at scale. Through IoH connectivity, humanoid robots could share anonymized patient-state representations, task execution policies, and operational constraints across wards or even between geographically distributed hospitals. This would enable dynamic task allocation (e.g., triage support and logistics routing), collaborative teleoperation assisted by fleet-level learning, and continuous improvement of medical procedures through shared experience. Such networked coordination transforms humanoids from isolated assistants into adaptive components of a distributed healthcare ecosystem, augmenting human clinicians rather than replacing them.

C. Education and Special Needs Support

Educational applications of humanoid robots are advancing rapidly with the integration of large language models and natural language processing. Recent deployments position humanoids such as Pepper as interactive teaching assistants capable of delivering multimodal instruction via speech, gesture, and visual displays [97]. Classroom studies report generally positive student reception, particularly in small-group and individualized learning settings.

Effective humanoid tutors must carefully align verbal explanations with nonverbal cues and pedagogical timing to ensure that embodiment enhances—rather than distracts from—learning, a requirement that is especially important for learners with special needs [98]. Within the IoH framework, educational robots can share anonymized learning analytics and teaching strategies across institutions, enabling continuous improvement and scalable personalization of robotic education.

D. Social Companionship and Emotional Support

Humanoid robots are increasingly explored for social companionship and emotional support. Beyond early therapeutic systems such as PARO, recent studies show that humanoid companions can reduce stress and loneliness through structured interaction and self-disclosure protocols [99].

Anthropomorphic features—such as facial expression, gaze, and posture—play a key role in establishing trust and emotional engagement. Within an IoH ecosystem, humanoid companions can share affective interaction models and dialog strategies, improving adaptability across users and cultures. At the same time, ethical concerns surrounding emotional dependency and autonomy highlight the need for transparent design and governance.

E. Household Assistance and Domestic Robots

Humanoid robots are increasingly capable of performing household tasks such as cleaning, serving, and object manipulation, as demonstrated by systems including Justin, EDAN, T-HR3, and Care-O-bot [100]. These platforms rely on modular designs that support adaptation to diverse home environments.

IoH significantly amplifies domestic assistance by enabling robots to share semantic maps, manipulation strategies, and anonymized user preferences across households. This collective learning paradigm transforms household robots from standalone appliances into continuously improving assistants while raising important challenges related to privacy and user trust.

F. Research and Embodied Cognition

Open-source humanoid platforms such as iCub play a central role in research on embodied cognition, enabling studies of learning, manipulation, and human–robot interaction through physical experience [101].

IoH-style connectivity allows research humanoids to share learned representations, experimental data, and control policies across laboratories, accelerating scientific progress and

improving reproducibility. Cost-effective platforms such as Salvius further democratize access to humanoid research [102], supporting large-scale collaborative experimentation within IoH ecosystems.

G. Industry Environments (Smart Factories)

In industrial settings, IoH-enabled humanoids act as flexible, networked coworkers integrated with existing IIoT infrastructure. Unlike fixed automation, humanoids can navigate human-designed workspaces, collaborate safely with workers, and adapt to new tasks with minimal reconfiguration [103].

Through IoH, industrial humanoids share task models, operational data, and learned policies across factories, enabling predictive maintenance, adaptive quality inspection, and dynamic scheduling. This layered integration supports local autonomy alongside global optimization, positioning humanoid coworkers as a unifying automation layer in smart manufacturing.

H. Public Engagement and Competitions

Events such as the 2025 World Humanoid Robot Games in Beijing provide standardized benchmarks for humanoid capabilities while raising public awareness [104]. Beyond competition, these events generate valuable datasets, foster international collaboration, and stimulate discussion on the societal impact of humanoid robots.

V. CHALLENGES

Although many challenges associated with IoH have been individually studied in IoT, networking, AI, and robotics, their manifestation in IoH is fundamentally different. This is because humanoid robots are embodied, force-exerting, autonomous agents operating in close physical and social proximity to humans. As a result, challenges that are tolerable or well-contained in traditional cyber systems can directly translate into physical harm, safety violations, or loss of trust in IoH. Moreover, the absence of standardized benchmarks and evaluation frameworks further complicates the assessment and comparison of IoH technologies, making it difficult to objectively measure progress or validate proposed solutions. In this section, we revisit these challenges and explicitly justify why existing solutions are insufficient when applied to large-scale, networked humanoid systems.

In this section, we discuss the challenges of IoH, including security and privacy, scalability, ethical and social implications, interoperability, and the lack of standardized benchmarks and evaluation frameworks.

A. Security and Privacy

The IoH expands the attack surface from a single robot to a globally connected fleet spanning homes, factories, and cloud/edge resources. While security threats such as unauthorized access, data leakage, and denial-of-service attacks are well studied in IoT and cloud systems, their consequences in IoH are substantially more severe because compromised humanoids can physically interact with humans

and the environment. In addition to classical cyber threats, humanoids introduce new attack vectors at the perception and decision layers, particularly through adversarial manipulation of learned policies and multimodal foundation models. Recent work shows that multitenant ROS2 deployments can be compromised via isolation escapes and information leakage—even with DDS-Security configured—through enclave or namespace bypass and cross-topic covert channels; mixed ROS1/ROS2 bridges can further reintroduce legacy weaknesses [105].

A critical humanoid-specific vulnerability arises from adversarial attacks on visual locomotion and manipulation policies. Modern humanoids increasingly rely on VLA models and LMMs to perceive terrain, recognize affordances, and plan motion. Recent studies demonstrate that carefully crafted visual perturbations—such as adversarial textures, sensor-level attacks, or environment-level modifications—can significantly degrade task success or even induce unsafe behaviors in robotic systems [106], [107]. Unlike image-classification attacks, these perturbations directly propagate into closed-loop control, resulting in unstable gaits, incorrect foot placement, or unsafe object interactions.

Beyond node compromise, securing data flows across heterogeneous subsystems is critical. PICAROS demonstrates decentralized information-flow control for ROS2 to constrain who can read, transform, or broadcast messages, providing a practical basis for least-privilege data exchange across computation graphs [108]. However, for humanoids that close tight perception–decision–action loops, adversarial inputs injected at the perception layer can bypass traditional access control entirely, propagating through learned policies before security monitors or middleware-level checks can intervene.

Hardening IoH middleware therefore requires a defense-in-depth program: 1) strict OS/container isolation per tenant and per safety-critical function; 2) least-privilege DDS-Security policies (SROS2 enclaves with minimal and audited permissions); 3) authenticated discovery and access control aligned with the OMG DDS-Security specification; and 4) systematic “graph mutation” testing and fuzzing before deployment [109], [110]. For LMM-driven humanoids, this program must be augmented with perception- and model-level defenses, including adversarial training against physically realizable perturbations, robustness evaluation under sensor attacks, and multimodal consistency checks that cross-validate vision, language, and proprioceptive signals [111], [112]. Formal studies corroborate that misconfiguration and uneven implementations remain frequent root causes [113], [114].

Recent work further suggests that system-level defenses are essential. Combining adversarially trained policies with run-time safety constraints and hierarchical control architectures can bound the physical consequences of adversarial decisions, even when upstream VLA or LMM components are partially compromised [111]. Such approaches are particularly critical for humanoid locomotion, where stability margins are narrow and recovery time is limited.

Privacy risks in IoH go beyond control integrity. Continuous capture of audio–video, biometrics, posture, and behavioral context makes data minimization essential. Moreover, LMM-

based perception pipelines may unintentionally infer sensitive attributes or behaviors not required for task execution, amplifying privacy risks beyond those of conventional IoT sensors. Unlike smart-home IoT devices, humanoids operate as mobile observers within intimate human spaces, significantly increasing the risk of unintended inference and long-term profiling. On-device redaction and anonymization pipelines, encryption in transit and at rest, and hardware-rooted identity with remote attestation are therefore necessary but still insufficient without humanoid-aware privacy policies [115]. Fog/cloud offloading frameworks (e.g., FogROS2 variants) must be carefully constrained to avoid creating privileged communication paths that bypass local safety checks [116].

Overall, existing IoT and cloud-security solutions cannot be directly reused for IoH because they do not account for adversarial robustness of learned perception and decision models, embodiment-specific safety constraints, or the tight coupling between multimodal inference and physical actuation.

B. Networking Constraints for Stabilizing High-DoF Humanoid Systems

A defining challenge unique to IoH lies in stabilizing high-DoF humanoid dynamics over shared and often wireless networks. Unlike conventional IoT nodes or even wheeled robots, humanoids rely on tightly coupled, high-frequency control loops for balance, locomotion, and manipulation, making them extremely sensitive to communication latency, jitter, and bandwidth fluctuations.

From a control-theoretic perspective, stable operation of a humanoid robot requires that the end-to-end delay of the control loop remains below a task-dependent bound. Let τ_{net} denote network-induced latency, τ_{comp} denote the computation delay (state estimation, control, and planning), and $\tau_{\text{ctrl}}^{\text{max}}$ denote the maximum admissible control-loop delay for stability. A necessary condition for stable closed-loop control can be expressed as

$$\tau_{\text{net}} + \tau_{\text{comp}} < \tau_{\text{ctrl}}^{\text{max}}. \quad (1)$$

For humanoid balance and whole-body locomotion, $\tau_{\text{ctrl}}^{\text{max}}$ is typically on the order of 1–5 ms, particularly for feedback loops involving the center of mass, zero-moment point, or contact forces. Delays exceeding this bound can result in loss of balance, unstable gaits, or unsafe contact with the environment.

Bandwidth consumption in IoH is similarly dominated by high-frequency sensor modalities. Let f_i denote the update frequency of sensor modality i and s_i denote its payload size per sample. The aggregate required bandwidth B can be approximated as

$$B = \sum_i f_i \cdot s_i. \quad (2)$$

For a typical humanoid platform, joint encoders (30–40 DoF) operate at 1–2 kHz, IMUs at ~ 1 kHz, and tactile skins at hundreds of hertz to 1 kHz, resulting in several megabits per second of continuous, latency-sensitive traffic even before considering vision or force–torque sensing. In contrast, high-bandwidth modalities such as stereo or RGB-D cameras

(30–60 Hz) can generate hundreds of megabits per second to gigabits per second of raw data, which cannot be streamed over wireless links without aggressive compression or offloading.

These quantitative constraints explain why existing cloud-robotics and IoT networking assumptions—where delays of tens of milliseconds and best-effort delivery are acceptable—do not hold for IoH. As a result, safety-critical control loops must remain local to the humanoid, while wireless networking (e.g., 5G/6G ultra-reliable low-latency communication (URLLC) or time-sensitive networking (TSN)-enabled links) is primarily leveraged for higher level coordination, perception sharing, and learning.

The lack of standardized benchmarks and evaluation frameworks further complicates this issue; there is currently no consensus on how to quantitatively evaluate the impact of latency, jitter, and packet loss on humanoid stability across platforms. Developing shared metrics, reference tasks, and network-in-the-loop testbeds is therefore a critical open research direction for IoH.

C. Scalability

Scaling the IoH requires handling telemetry, perception, and control data from thousands of humanoid robots—an infrastructure challenge closer to cloud-scale systems than classical robotics. While scalable data ingestion and stream processing have been extensively studied in distributed systems, humanoid fleets generate high-frequency, tightly synchronized control and perception streams whose timing directly affects physical stability and safety. High-frequency proprioceptive and multimodal sensor data must therefore be processed under strict latency and jitter constraints [117].

Hybrid edge–cloud architectures are commonly proposed to reduce network load, keeping safety-critical control onboard while offloading perception and analytics to edge or cloud servers [118]. However, existing cloud-scaling techniques assume that delayed or dropped packets degrade performance gracefully, an assumption that does not hold for humanoid locomotion, manipulation, or physical collaboration. Even brief overloads or scheduling delays can destabilize whole-body control loops.

At the middleware level, ROS2 and DDS offer flexible publish/subscribe semantics, but scaling to large humanoid fleets is constrained by participant limits, discovery overhead, and cross-domain bridging costs [119]. Unlike sensor networks, humanoid systems require synchronized, cross-topic consistency across perception, planning, and actuation, which current middleware was not originally designed to guarantee. Deterministic networking (e.g., TSN) offers promise, but integration with wireless and mobile humanoid platforms remains an open challenge [120].

D. Ethical and Social Implications

The IoH raises ethical and social concerns that extend beyond those of conventional automation and IoT systems. While job displacement and privacy are well-studied in AI and automation, humanoid robots introduce additional risks due to their human-like form, social presence, and ability to

form long-term interactions. As humanoids gain autonomy and task competence, concerns about labor displacement are compounded by issues of dependency, emotional attachment, and erosion of human-to-human care [121], [122].

Autonomy and values alignment are particularly challenging in sensitive settings such as healthcare, elder care, and education. Unlike decision-support AI systems, humanoids act continuously and physically in social contexts, making errors or misaligned behaviors more visible and potentially more harmful. Cross-cultural studies show that acceptance and trust in humanoid systems vary widely, underscoring the need for context-aware deployment, transparent behavior, and participatory design [123], [124].

E. Interoperability

Interoperability is a central challenge for IoH. While interoperability has long been studied in IoT and industrial automation, humanoid interoperability must span not only data formats and communication protocols, but also embodiment, control abstractions, semantic understanding, and evaluation criteria. Proprietary hardware and software stacks hinder collaboration across humanoid platforms and integration with IoT and cloud systems [125].

ROS2 provides a common middleware foundation, but vendor-specific DDS implementations, legacy APIs, and inconsistent QoS configurations limit seamless interoperability [126]. More importantly, existing interoperability solutions focus on syntactic compatibility, whereas IoH requires semantic interoperability and shared evaluation frameworks—i.e., common benchmarks, metrics, and test scenarios—to meaningfully compare behaviors, safety, and performance across humanoid systems. Emerging protocols such as A2A and MCP provide promising mechanisms for cross-vendor coordination, but governance, standardization, and benchmark definition remain open research challenges.

In summary, although IoH challenges resemble those studied in IoT, networking, AI, and robotics, their coupling with embodiment, autonomy, and human interaction fundamentally changes their nature. The lack of standardized benchmarks and evaluation frameworks further limits objective comparison and quantitative analysis, underscoring the need for IoH-specific metrics, datasets, and testbeds. Existing solutions cannot be directly applied without rethinking assumptions about latency tolerance, safety boundaries, semantic consistency, accountability, and evaluation methodology.

VI. FUTURE RESEARCH DIRECTIONS

The IoH remains in its infancy, and many open problems must be addressed before large-scale, safe, interoperable, and measurably comparable humanoid networks can be realized. Building on the challenges outlined earlier, we identify six major research directions that are emerging in the literature. Across all directions, a recurring limitation is the lack of standardized benchmarks, metrics, and evaluation frameworks, which hinders objective comparison, reproducibility, and quantitative assessment of competing IoH approaches. Table VIII summarizes these research directions.

TABLE VIII
FUTURE RESEARCH DIRECTIONS FOR THE IoH

Research Direction	Role in IoH	Representative Systems / Examples	References
Secure-by-Design Middleware & Policy Verification	Foundational to trustworthy IoH; addresses isolation escapes, covert channels, and vendor misconfigurations in ROS2/DDS. Future work must also define security benchmarks and adversarial evaluation suites to quantitatively assess middleware robustness at fleet scale.	PICAROS (policy-aware DIFC); SGX-based DDS enclaves with remote attestation; formal fuzzing/verification of ROS2 graphs	[113], [127]
Scalable Cloud-Edge Scheduling & Observability	Balances real-time control on humanoids with cloud/edge offloading for perception and analytics. Lack of standardized latency, jitter, and failure-recovery metrics makes it difficult to compare orchestration strategies across platforms.	FogROS2 for latency-aware scheduling; policy-based routing; observability/tracing frameworks for distributed ROS2	[87]
Deterministic Networking	Enables sub-ms closed-loop actuation and bounded jitter for humanoid fleets. Evaluation frameworks that jointly measure control stability, safety margins, and network behavior under mobility remain largely unexplored.	DDS/ROS2 with TSN backbones; wireless TSN research for mobile robots; scheduler co-design for fleet coordination	[120]
Fleet-Scale Learning, Personalization & Privacy	Collective fleet learning promises generalization and personalization across humanoids. Benchmarking privacy-utility trade-offs, personalization gains, and robustness to poisoned or heterogeneous data is an open problem.	Open X-Embodiment dataset; federated learning for HRC; blockchain-secured FL; Internet of Intelligence	[128]–[129]
Humanoid Loco-Manipulation & Benchmarking	Core to real-world deployment; whole-body loco-manipulation remains unsolved. This direction is central to establishing standardized, reproducible benchmarks for humanoid control and IoH-scale coordination.	Humanoid-Gym; HumanoidBench; BEHAVIOR-1K; DexCap data pipelines	[20], [21], [133]
Interoperability & Semantic Standards	Persistent barrier for IoH-scale integration. Shared semantic representations and evaluation criteria are required to meaningfully compare interoperability solutions across vendors and domains.	ROS2 middleware; Zenoh bridging; micro-ROS; Agent-to-Agent (A2A) and Model Context Protocol (MCP)	[126]

A. Secure-by-Design Middleware and Policy Verification

Security remains the foundational concern for IoH. Recent work has shown that multitenant ROS2 deployments can be compromised through isolation escapes and cross-topic covert channels, despite DDS-Security being enabled [113]. Decentralized information-flow control (DIFC) approaches such as PICAROS introduce fine-grained policies for who can read, transform, or broadcast messages [127].

Beyond middleware hardening, secure-by-design IoH architectures must explicitly incorporate adversarial robustness guarantees for VLA- and LMM-driven perception and control. This includes standardized threat models covering physical-world adversarial attacks, benchmark suites for evaluating robustness of visual locomotion and manipulation policies, and integration with secure federated learning mechanisms that defend against poisoning and inference attacks in fleet-scale training [106], [107], [134]. Hardware-based identity and trusted execution (e.g., DDS inside SGX enclaves with remote attestation) remain promising directions for securely onboarding humanoid nodes into fleet graphs.

B. Scalable Cloud-Edge Scheduling and Observability

Scaling humanoid fleets requires hybrid cloud-edge orchestration, where safety-critical control remains on-robot, while compute-intensive tasks are offloaded to edge and cloud backends. Systems such as FogROS2 [87] demonstrate latency-aware scheduling, yet predictable real-time performance remains elusive.

A key research gap lies in the absence of unified observability metrics and evaluation frameworks that jointly capture end-to-end latency, control stability, fault recovery, and energy

efficiency across heterogeneous infrastructures. Addressing this gap is essential for an objective comparison of orchestration strategies.

C. Deterministic Networking for Humanoid Fleets

IoH introduces stringent networking requirements, with closed-loop humanoid actuation often demanding submillisecond latency and bounded jitter. TSN has been explored as a backbone for DDS-based middleware [120], but integration with wireless 5G/6G and mobile humanoid nodes remains unresolved.

Future research should establish evaluation methodologies that correlate network-level guarantees with physical safety, control performance, and degradation behavior under mobility and packet loss.

D. Fleet-Scale Learning, Personalization, and Privacy

A central promise of IoH is collective improvement through shared learning. Cross-embodiment datasets such as Open X-Embodiment [128] enable generalization, while federated learning supports privacy-preserving collaboration [129].

However, there is currently no consensus on benchmarks for evaluating personalization quality, privacy leakage, robustness to adversarial updates, or long-term fleet-level learning stability. Developing standardized evaluation protocols remains a critical open problem.

E. Humanoid Loco-Manipulation and Benchmarking

IoH will only be impactful if humanoids can reliably perform locomotion and manipulation in diverse environments.

Progress in locomotion and manipulation is evident [20], [21], but whole-body loco-manipulation remains challenging.

Existing benchmarks focus primarily on single-robot performance; extending them to multihumanoid, networked, and human-in-the-loop scenarios is essential for IoH-relevant evaluation.

F. Interoperability and Semantic Standards

Interoperability remains a persistent barrier. While ROS2 provides a common middleware, vendor-specific DDS implementations hinder seamless collaboration [126].

Future work must go beyond protocol compatibility to define shared semantic models and evaluation criteria—i.e., how well humanoids align on goals, safety constraints, and context—which are prerequisites for quantitative comparison of interoperability solutions. Emerging protocols such as A2A and MCP may help address these challenges.

VII. CONCLUSION AND FUTURE WORK

The IoH represents a unifying paradigm that combines advanced AI, next-generation connectivity, and distributed systems to enable collaborative, autonomous humanoid networks. This survey reviewed key enabling technologies—including LMMs, reinforcement and IL, advanced sensing and actuation, and emerging communication protocols such as A2A and MCP—along with representative applications in healthcare, elder care, industry, and education. Despite its promise, IoH faces substantial challenges related to security and privacy, scalability, interoperability, ethical deployment, and real-time coordination. Addressing these issues requires future research on secure-by-design middleware, predictable cloud-edge scheduling, deterministic and low-latency networking, and privacy-preserving fleet-scale learning. Equally important is the development of standardized semantic frameworks, benchmarks, and evaluation methodologies for humanoid loco-manipulation and collaborative behaviors. By advancing these directions through interdisciplinary efforts, IoH can mature into a scalable, trustworthy, and impactful ecosystem, fundamentally redefining human-robot collaboration in real-world environments.

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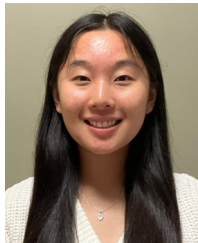
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